$\because \frac{\partial Y}{\partial X} = \frac{\partial Y_S}{\partial X} + \frac{\partial Y_F}{\partial X} = J_S(X) + J_F(X)$

$\therefore \frac{d_v L}{dX} = J_S^T \frac{d_v L}{dY} + J_F^T \frac{d_v L}{dY}$

$\frac{dL}{dX} = \left(\frac{d_v L}{dX}\right).\text{reshape}(X.\text{shape})$

here $J_S(X) = \begin{pmatrix} \frac{\partial Y_{S1}}{\partial X_1} & & \frac{\partial Y_{S1}}{\partial X_i} & & \frac{\partial Y_{S1}}{\partial X_m} \end{pmatrix}$

$= \text{diag}\left(\frac{\partial Y_{S1}}{\partial X_1}, \frac{\partial Y_{Si}}{\partial X_i}, \frac{\partial Y_{Sm}}{\partial X_m}\right)$

$J_F(X) = \text{diag}\left(\frac{\partial Y_{F1}}{\partial X_1}, \frac{\partial Y_{Fi}}{\partial X_i}, \cdots\right)$

② (method 2)

$\because \left(\frac{dL}{dX}\right)_{:,i} = J_S^T(X_i)\left(\frac{dL}{dY}\right)_{:,i} + J_F^T(X_i)\left(\frac{dL}{dY}\right)_{:,i}$ (from 2)

$\therefore \begin{pmatrix} \frac{dL}{dX_1} \\ \frac{dL}{dX_m} \end{pmatrix} = \begin{pmatrix} \left(\frac{\partial Y_{S1}}{\partial X_1}\right)^T & & \left(\frac{\partial Y_{F1}}{\partial X_1}\right)^T \\ \left(\frac{\partial Y_{Sm}}{\partial X_m}\right)^T & & \left(\frac{\partial Y_{Fm}}{\partial X_m}\right)^T \end{pmatrix} \begin{pmatrix} \frac{dL}{dY_1} \\ \frac{dL}{dY_m} \end{pmatrix}$

$\therefore \frac{d_v L}{dX} = J_S^T \frac{d_v L}{dY} + J_F^T \frac{d_v L}{dY}$

③ $\frac{dL}{d\theta_S} = \sum_{i=1}^{m} \left(\frac{\partial Y_{Si}}{\partial \theta_S}\right)^T \frac{dL}{dY_i}$

$= \left(\left(\frac{\partial Y_{S1}}{\partial \theta_S}\right)^T, \cdots, \left(\frac{\partial Y_{Sm}}{\partial \theta_S}\right)^T\right) \frac{d_v L}{dY}$,

④ $\frac{dL}{d\theta_F} = \sum_{i=1}^{m} \left(\frac{\partial Y_{Fi}}{\partial \theta_F}\right)^T \frac{dL}{dY_i}$

③ $\because \frac{\partial L}{\partial \theta_S} = \frac{\partial L}{\partial Y} \frac{\partial Y}{\partial \theta_S}$

$\therefore \frac{dL}{d\theta_S} = \left(\frac{\partial Y}{\partial \theta_S}\right)^T \frac{dL}{dY} = \left(\frac{\partial Y_S}{\partial \theta_S}\right)^T \frac{dL}{dY}$

④ $\frac{dL}{d\theta_F} = \left(\frac{\partial Y_F}{\partial \theta_F}\right)^T \frac{dL}{dY}$

3. $Y = B(X)$, $X = (X_1, X_i, X_m) \in R^{d_{in} \times m}$

$Y = (Y_1, Y_i, Y_m) \in R^{d_{out} \times m}$

3.2 $Y_i = B(X_i) = S(X_i; \theta_S) + F(X_i; \theta_F)$

3.3 $Y = B(X) = Y_S + Y_F$,

$Y_S = S(X; \theta_S)$, $Y_F = F(X; \theta_F)$

$Y_S = (Y_{S1}, Y_{Si}, Y_{Sm})$

$Y_F = (Y_{F1}, Y_{Fi}, Y_{Fm})$

$Y_{Si} = S(X_i; \theta_S)$, $Y_{Fi} = F(X_i; \theta_F)$

4. $L = L(Y) \in R$, $\frac{dL}{dY} \Rightarrow \frac{dL}{dX}, \frac{dL}{d\theta_S}, \frac{dL}{d\theta_F}$

4.1 $X.\text{vec} := \begin{pmatrix} X_1 \\ X_i \\ X_m \end{pmatrix}$ if $X = (X_1, X_i, X_m)$

$\frac{dy}{d} := \frac{d().\text{vec}}{d().\text{vec}}$, $\frac{\partial()}{\partial} := \frac{\partial().\text{vec}}{\partial().\text{vec}}$

4.2 ① (method 1)

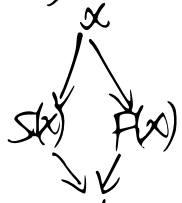
$\because \frac{\partial_v L}{\partial X} = \frac{\partial_v L}{\partial Y} \frac{\partial_v Y}{\partial X}$ $\therefore \frac{d_v L}{dX} = \left(\frac{\partial_v Y}{\partial X}\right)^T \frac{d_v L}{dY}$

residual block

1. residual block: $y = B(x)$, $x \in R^{d_{in}}$, $y \in R^{d_{out}}$

1.2 $y = B(x) = y_S + y_F$, $y_S = S(x; \theta_S)$, $y_F = F(x; \theta_F)$

here $S(x) = x$ if $d_{in} = d_{out}$



1.3 $S(x)$: shortcut function.

$F(x)$: residual function.

2. $L = L(y) \in R$, $\frac{dL}{dy} \Rightarrow \frac{dL}{dx}, \frac{dL}{d\theta_S}, \frac{dL}{d\theta_F}$.

($\frac{dL}{dX} := \left(\frac{\partial L}{\partial X}\right).\text{reshape}(X.\text{shape}), \forall X$).

solve: ① $\frac{\partial y}{\partial x} = \frac{\partial y_S(x)}{\partial x} + \frac{\partial y_F}{\partial x}$

$\frac{\partial y}{\partial \theta_S} = \frac{\partial y_S}{\partial \theta_S}$, $\frac{\partial y}{\partial \theta_F} = \frac{\partial y_F}{\partial \theta_F}$

② $\because \frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial x}$

$\therefore \frac{dL}{dx} = \left(\frac{\partial y}{\partial x}\right)^T \frac{dL}{dy} = \left(\frac{\partial y_S}{\partial x}\right)^T \frac{dL}{dy} + \left(\frac{\partial y_F}{\partial x}\right)^T \frac{dL}{dy}$

$= J_S^T(x) \frac{dL}{dy} + J_F^T(x) \frac{dL}{dy}$

($= \frac{dL}{dy} + J_F^T(x) \frac{dL}{dy}$, if $S(x) = x$)